

LAB 1

20CP207P – Operating Systems Lab

Title:

Introduction of basic Linux commands.

Task:

Perform the basic Linux commands with all the options available for them like p, f, a, t, r, S, R, i, g, h, d etc. and understand how it works'

Commands:

- `ls`

Lists files and directories in the current directory.

Options:

- `-a`: Includes hidden files (those starting with `.`) in the listing.
- `-l`: Displays detailed information (permissions, owner, size, etc.) in a long format.
- `-h`: Shows file sizes in a human-readable format (e.g., KB, MB).
- `-R`: Recursively lists contents of subdirectories.
- `-t`: Sorts files by modification time, with the newest first.
- `-S`: Sorts files by size, with the largest first.
- `-r`: Reverses the sorting order.
- `-1`: Lists one file per line.
- `-color`: Highlights file types and permissions using colors.

```
labex:lab_1_os/ $ ls -l -a -R -S
.:
total 4
-rw-rw-r-- 1 labex labex 68 Jan 13 00:02 my_c.c
drwxrwxr-x 3 labex labex 65 Jan 13 00:21 .
drwxrwxr-x 2 labex labex 39 Jan 13 00:22 nesteddir
drwxr-xr-x 1 labex labex 22 Jan 13 00:00 ..
-rw-rw-r-- 1 labex labex  0 Jan 13 00:03 myc2.c
-rw-rw-r-- 1 labex labex  0 Jan 13 00:03 myc3.c

./nesteddir:
total 4
drwxrwxr-x 3 labex labex 65 Jan 13 00:21 ..
drwxrwxr-x 2 labex labex 39 Jan 13 00:22 .
-rw-rw-r-- 1 labex labex  1 Jan 13 00:22 myc4.c.save
-rw-rw-r-- 1 labex labex  0 Jan 13 00:22 myc4.c
labex:lab_1_os/ $
```

- `cd`

Changes the current working directory.

Options:

- `cd ..`: Moves to the parent directory.
- `cd ~`: Changes to the home directory.
- `cd -`: Switches to the previous working directory.
- `cd /path/to/directory`: Navigates to the specified directory.

```
File Edit View Terminal Tabs Help
labex:lab_1_os/ $ cd nesteddir
labex:nesteddir/ $ ls
myc4.c  myc4.c.save
labex:nesteddir/ $
```

```
File Edit View Terminal Tabs Help
labex:lab_1_os/ $ cd nesteddir
labex:nesteddir/ $ ls
myc4.c  myc4.c.save
labex:nesteddir/ $ cd ~
labex:~/ $
```

- `pwd`

Prints the full path of the current working directory

Options:

- `-L`: Displays the logical current directory, including symbolic links (default behavior).
- `-P`: Displays the physical current directory by resolving all symbolic links.

```
File Edit View Terminal Tabs Help
labex:lab_1_os/ $ pwd
/home/labex/project/lab_1_os
labex:lab_1_os/ $ pwd -L
/home/labex/project/lab_1_os
labex:lab_1_os/ $ pwd -P
/home/labex/project/lab_1_os
labex:lab_1_os/ $
```

- `mkdir`

Creates a new directory.

Options:

- **-p**: Creates parent directories as needed (doesn't return an error if they already exist).
- **-v**: Displays a message for each created directory (verbose mode). (Verbose mode is a setting that provides detailed information about what a system, program, or process is doing.)
- **-m <mode>**: Sets permissions (in octal format) for the new directory.

```
labex:lab_1_os/ $ pwd
/home/labex/project/lab_1_os
labex:lab_1_os/ $ pwd -L
/home/labex/project/lab_1_os
labex:lab_1_os/ $ pwd -P
/home/labex/project/lab_1_os
labex:lab_1_os/ $ mkdir somenewdir
labex:lab_1_os/ $ mkdir -v onedir
mkdir: created directory 'onedir'
labex:lab_1_os/ $ mkdir -m 755 twodir
labex:lab_1_os/ $ ls
myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir
labex:lab_1_os/ $
```

- **rm**

Removes files or directories.

Options:

- **-f**: Forces deletion without prompting for confirmation.
- **-r**: Recursively deletes directories and their contents.
- **-i**: Prompts for confirmation before deleting each file.
- **-v**: Displays detailed information about the deletion process (verbose mode).

```
labex:lab_1_os/ $ ls
del1.c del2.c myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir
labex:lab_1_os/ $ rm -f del1.c
labex:lab_1_os/ $ ls
del2.c myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir
labex:lab_1_os/ $ rm -i del2.c
rm: remove regular empty file 'del2.c'? y
labex:lab_1_os/ $ ls
myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir
labex:lab_1_os/ $
```

- `cp`

Copies files and directories.

Options:

- `-u`: Preserves file attributes like timestamps, permissions, and ownership.
- `-p`: Copies only if the source file is newer than the destination or if the destination file doesn't exist.

```
labex:lab_1_os/ $ ls
myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir
labex:lab_1_os/ $ cp my_c.c nesteddir/c123.c
labex:lab_1_os/ $ cd nesteddir
labex:nesteddir/ $ ls
c123.c myc4.c myc4.c.save
labex:nesteddir/ $ cd ..
labex:lab_1_os/ $ cp -p my_c.c nesteddir/c234.c
labex:lab_1_os/ $ ls -R
.:
myc2.c myc3.c my_c.c nesteddir onedir somenewdir twodir

./nesteddir:
c123.c c234.c myc4.c myc4.c.save

./onedir:

./somenewdir:

./twodir:
labex:lab_1_os/ $
```

- `mv`

Moves or renames files and directories.

```

labex:lab_1_os/ $ mv myc3.c onedir/myc3.c
labex:lab_1_os/ $ ls -R
.:
myc2.c  my_c.c  nesteddir  onedir  somenewdir  twodir

./nesteddir:
c123.c  c234.c  myc4.c  myc4.c.save

./onedir:
myc3.c

./somenewdir:

./twodir:
labex:lab_1_os/ $

```

- `touch`

Creates an empty file or updates the timestamp of an existing file.

```

labex:project/ $ touch hello.c
labex:project/ $ ls
hello.c  lab1os  myc1.c  myc2.c
labex:project/ $

```

- `cat`

Concatenates and displays the contents of one or more files.

```

labex:project/ $ touch hello.c
labex:project/ $ ls
hello.c  lab1os  myc1.c  myc2.c
labex:project/ $ cat hello.c
#include <stdio.h>

int main (void)
{
    printf("Lab 1 OS !")
}
labex:project/ $

```

- nano

A simple command-line text editor for creating and editing files.

```

GNU nano 6.2                                hello.c
#include <stdio.h>

int main (void)
{
    printf("Lab 1 OS !")
}

[ Read 6 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
  
```

- grep

Searches for patterns within files and outputs matching lines.

```

labex:project/ $ ls
hello.c lablos myc1.c myc2.c
labex:project/ $ grep "include" helloc.c
grep: helloc.c: No such file or directory
labex:project/ $ grep "include" hello.c
#include <stdio.h>
labex:project/ $
  
```

- sudo

Executes a command with superuser (root) privileges or as another user.

```

labex:project/ $ sudo
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABknS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABknS] [-g group] [-h host] [-p prompt] [-U user] [-u user]
        [command]
usage: sudo [-ABbEHknPS] [-r role] [-t type] [-C num] [-D directory] [-g group]
        [-h host] [-p prompt] [-R directory] [-T timeout] [-u user]
        [VAR=value] [-i|-s] [<command>]
usage: sudo -e [-ABknS] [-r role] [-t type] [-C num] [-D directory] [-g group]
        [-h host] [-p prompt] [-R directory] [-T timeout] [-u user] file ...
labex:project/ $

```

- **apt-get**

A package management command-line tool for handling packages in Debian-based distributions (like Ubuntu).

Options:

- **install** <package>: Installs the specified package.
- **remove** <package>: Removes the specified package.
- **update**: Updates the package list from the repositories.
- **upgrade**: Upgrades all installed packages to the latest version.

```

labex:project/ $ apt-get
apt 2.4.12 (amd64)
Usage: apt-get [options] command
        apt-get [options] install|remove pkg1 [pkg2 ...]
        apt-get [options] source pkg1 [pkg2 ...]

apt-get is a command line interface for retrieval of packages
and information about them from authenticated sources and
for installation, upgrade and removal of packages together
with their dependencies.

```

- **dpkg**

A low-level package manager for Debian-based distributions, used for installing, building, removing, and querying .deb packages.

- **chmod**

Changes the file or directory permissions.

```

labex:project/ $ chmod --help
Usage: chmod [OPTION]... MODE[,MODE]... FILE...
       or: chmod [OPTION]... OCTAL-MODE FILE...
       or: chmod [OPTION]... --reference=RFILE FILE...
Change the mode of each FILE to MODE.
With --reference, change the mode of each FILE to that of RFILE.

  -c, --changes          like verbose but report only when a change is made
  -f, --silent, --quiet  suppress most error messages
  -v, --verbose          output a diagnostic for every file processed
                        --no-preserve-root  do not treat '/' specially (the default)
                        --preserve-root    fail to operate recursively on '/'
                        --reference=RFILE  use RFILE's mode instead of MODE values
  -R, --recursive       change files and directories recursively
  --help                display this help and exit
  --version              output version information and exit

Each MODE is of the form '[ugoa]*([-+=]([rwxXst]*|[ugo]))+|[-+=][0-7]+'.
```

- **chown**

Changes the ownership of a file or directory.

```

labex:project/ $ chown --help
Usage: chown [OPTION]... [OWNER][:[GROUP]] FILE...
       or: chown [OPTION]... --reference=RFILE FILE...
Change the owner and/or group of each FILE to OWNER and/or GROUP.
With --reference, change the owner and group of each FILE to those of RFILE.

  -c, --changes          like verbose but report only when a change is made
  -f, --silent, --quiet  suppress most error messages
  -v, --verbose          output a diagnostic for every file processed
                        --dereference      affect the referent of each symbolic link (this is
                        the default), rather than the symbolic link itself
  -h, --no-dereference   affect symbolic links instead of any referenced file
                        (useful only on systems that can change the
                        ownership of a symlink)
  --from=CURRENT_OWNER:CURRENT_GROUP
```

- **tar**

Archives and compresses files into a single file (tarball) or extracts them.

- **uname**

Displays system information such as the operating system name, kernel version, and other details.


```

labex:project/ $ uname
Linux
labex:project/ $ uname -s
Linux
labex:project/ $ uname -n
6783f510098fa2ceb37f7034
labex:project/ $ uname -r
5.4.0-162-generic
labex:project/ $ uname -v
#179-Ubuntu SMP Mon Aug 14 08:51:31 UTC 2023
labex:project/ $ uname -m
x86_64
labex:project/ $ uname -p
x86_64
labex:project/ $ uname -i
x86_64
labex:project/ $ uname -o
GNU/Linux
labex:project/ $ uname -a
Linux 6783f510098fa2ceb37f7034 5.4.0-162-generic #179-Ubuntu SMP Mon Aug 14 08:51:31 UTC 2023 x86_64 x86_64 GNU/Linux
labex:project/ $

```

- **date**

Displays or sets the system date and time.

```

labex:project/ $ date
Mon Jan 13 01:42:42 AM CST 2025
labex:project/ $

```

- **shutdown**

Shuts down or reboots the system in a controlled manner.

- **ps**

Displays information about currently running processes.

```

labex:project/ $ ps
  PID TTY          TIME CMD
 1297 pts/0        00:00:01 zsh
 2968 pts/0        00:00:00 ps
labex:project/ $

```

- `top`

Displays a dynamic, real-time view of the system's resource usage, including CPU, memory, and processes.

```
top - 01:49:01 up 83 days, 11:37, 0 users, load average: 0.25, 0.10, 0.11
Tasks: 43 total, 1 running, 42 sleeping, 0 stopped, 0 zombie
%Cpu(s): 2.1 us, 0.6 sy, 0.0 ni, 97.2 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 15728.3 total, 5947.7 free, 1917.8 used, 7862.8 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 13451.6 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
37	labex	20	0	834132	114780	64680	S	5.7	0.7	0:52.06	Xvnc
22	root	20	0	40824	28100	10652	S	0.3	0.2	0:00.58	supervisord
87	labex	20	0	468340	44152	32936	S	0.3	0.3	0:00.51	xfwm4
102	labex	20	0	227984	24848	19560	S	0.3	0.2	0:00.15	xfsettingsd
1270	labex	20	0	462116	38632	30148	S	0.3	0.2	0:01.11	xfce4-terminal
1	root	20	0	11204	3820	3540	S	0.0	0.0	0:00.01	init.sh
23	root	20	0	15424	9284	7652	S	0.0	0.1	0:00.01	sshd
24	root	20	0	14040	4400	3856	S	0.0	0.0	0:00.00	su
25	labex	20	0	40324	30808	6740	S	0.0	0.2	0:00.53	vncserver
47	labex	20	0	11204	3556	3308	S	0.0	0.0	0:00.00	sh
48	labex	20	0	11204	1880	1616	S	0.0	0.0	0:00.00	sh
49	labex	20	0	454068	76268	60472	S	0.0	0.5	0:00.22	xfce4-session
58	labex	20	0	8300	2020	1560	S	0.0	0.0	0:00.00	dbus-launch
59	labex	20	0	8624	3344	2676	S	0.0	0.0	0:00.08	dbus-daemon
61	labex	20	0	309460	7544	6884	S	0.0	0.0	0:00.00	at-spi-bus-laun
66	labex	20	0	8424	4532	4068	S	0.0	0.0	0:00.01	dbus-daemon
70	labex	20	0	231000	6296	5492	S	0.0	0.0	0:00.05	xfconfd
76	labex	20	0	162748	8016	7248	S	0.0	0.0	0:00.07	at-spi2-registr
81	labex	20	0	7972	1076	0	S	0.0	0.0	0:00.00	ssh-agent

- `du`

Displays the disk usage of files and directories.

```
labex:project/ $ du
0      ./lablos
8      .
labex:project/ $
```

- `df`

Displays information about disk space usage for file systems.

```

labex:project/ $ du
0      ./lablos
8      .
labex:project/ $ df
Filesystem      1K-blocks      Used Available Use% Mounted on
overlay         20971520    148976   20822544   1% /
tmpfs            65536         0      65536    0% /dev
tmpfs           8052884         0    8052884   0% /sys/fs/cgroup
shm              65536        7200     58336   11% /dev/shm
/dev/vdb         104806400  18582420  86223980  18% /etc/hosts
tmpfs            8052884         0    8052884   0% /proc/acpi
tmpfs            8052884         0    8052884   0% /proc/scsi
tmpfs            8052884         0    8052884   0% /sys/firmware
labex:project/ $ ls
hello.c lablos myc1.c myc2.c
labex:project/ $

```

- `history`

Displays a list of previously executed commands in the terminal session.

```

labex:project/ $ ls
hello.c lablos myc1.c myc2.c
labex:project/ $ history
 2  mkdir lablos
 3  touch myc1.c
 4  touch myc2.c
 6  touch hello.c
 8  cat hello.c
 9  nano hello.c
11  grep "include" helloc.c

```

- `ifconfig`

Displays or configures network interfaces, showing information about active network interfaces on the system.

```

labex:project/ $ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.19.0.4 netmask 255.255.0.0 broadcast 172.19.255.255
    ether 02:42:ac:13:00:04 txqueuelen 0 (Ethernet)
    RX packets 49219 bytes 4066301 (4.0 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 45652 bytes 24488842 (24.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 96 bytes 15082 (15.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 96 bytes 15082 (15.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

labex:project/ $

```

- `echo`

Displays a line of text or a variable's value to the terminal.

```

labex:project/ $ echo "I can do this all day"
I can do this all day
labex:project/ $

```

- `cal`

Displays a calendar for the current month or a specified month and year.

- `passwd`

Changes the password for a user account.

- `free`

Displays the system's memory usage, including total, used, free, and available memory.

- `comm`

Compares two sorted files line by line and outputs three columns: lines unique to the first file, lines unique to the second file, and lines common to both files.

- `group`

Displays the groups to which a user belongs.